Prevent Invalid Mining Rewards

Relevant files: `tx.py`, `blockchain.py`, `config.py`

Scenario

The UAP Coin blockchain creates a coinbase transaction whenever a block is mined.

Currently, the reward amount is defined by `CURRENT_REWARD`, but the blockchain does not independently verify whether the coinbase transaction is valid.

Task

Implement a method `validate_coinbase(tx, block_height)` that validates a newly generated coinbase transaction before it can be included in a block.

The method must check:

1. The transaction has exactly one input.
2. The input references `ZERO_HASH_BYTES` and index `0xFFFFFFFF`.
3. The coinbase input's script contains the expected block height in little-endian format.
4. The transaction has at least one output.
5. Every output amount is positive, and the combined output amount equals `CURRENT_REWARD * ONE_COIN`.

Integrate validation into `add_block()` so an invalid coinbase transaction is rejected before mining or writing to disk.

Expected demonstration

With a configured mining reward of 50 UAP Coin:

Test case	Expected result
One output containing 50 coins	Valid
Two outputs containing 30 and 20 coins	Valid
One output containing 60 coins	Invalid
One output containing −5 coins	Invalid
Coinbase referencing an ordinary transaction instead of the zero hash	Invalid

Use transaction objects constructed from the provided classes for testing. Students do not need to implement transaction signature verification.

Submission: **Submit only the blockchain.py file**